Graphic Era HILL UNIVERSITY
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University under section 2(f) of UGC Act, 1956

# End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: B.Tech ECE 3rd Semester
Name of the Paper: Electronic Devices
Paper Code: TEC-301

Time: 3 hour | Maximum Marks: 100

**Note:**

*(i)* All the questions are compulsory.
*(ii)* Answer any two sub questions from a, b and c in each main question.
*(iii)* Total marks for each question is 20 (twenty).
*(iv)* Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

a. Explain mobility and conductivity. Derive the expression for barrier potential in P-N junction diode. (CO1)

b. Explain energy band diagram for intrinsic and extrinsic semiconductor. Define diffusion and drift current. (CO1)

c. A silicon sample is doped with $5*10^{16}/cm^3$ phosphorus atoms. The mobility of electrons in silicon is $\mu_e$=1350 $cm^2$/v-s, and the mobility of holes is $\mu_h$=480 $cm^2$/v-s. Assume intrinsic carrier concentration $n_i$= $5*10^{16}/cm^3$ and elementary charge q=$1.6*10^{19}$ C. (CO1)

Find the conductivity of the sample.
How does the conductivity change if it is instead doped with $5*10^{16}/cm^3$ boron atoms.

Q2. (2X10=20 Marks)

a. Explain Tunnel diode and its negative resistance region characteristics with the help of diagram. (CO2)

b. Explain (CO2)
Light Emitting Diode
Zener Diode

c. Explain (CO2)
Varactor Diode
Photo Diode

Q3. (2X10=20 Marks)

a. Explain BJT as an amplifier (CO3)
What are the three regions of operation for a BJT, and how are they defined in terms of junction biasing?

b. (CO3)
Explain BJT internal capacitances.
Explain BJT T model.

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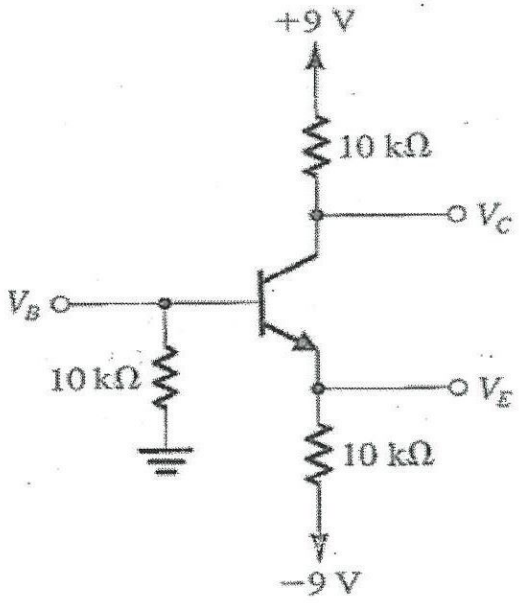
c. For the given circuit $V_B$= -1.5 V, assume $V_{BE}$=0.7 V. (CO3)
Calculate $V_E$, α, β and $V_C$. If a transistor with β=∞ is used, what values of $V_B$, $V_C$, and $V_E$ results?




Q4. (2X10=20 Marks)

a. Explain high frequency model of Common Source MOSFET. (CO4)

b. Derive relation between current and voltage for MOSFET. (CO4)

c. Draw PMOS and NMOS Voltage and Current characteristics for both enhancement and depletion type MOSFET. (CO4)

Q5. (2X10=20 Marks)

a. Explain (CO5)
(i) Photoresist
(ii) Oxidation

b. Explain CZ process for wafer preparation in detail. (CO5)

c.
Explain (CO5)
(i) Photolithography
(ii) Etching